

Quiz day 5

1. Consider the recurrence equation $y_n - 9y_{n-1} + 18y_{n-2} = 0$.

Find all solutions of the form $y_n = \lambda^n$.

2. You know that $r_n = c \cdot 3^n + d \cdot n \cdot 3^n + n$ is the solution of some recurrence equation with initial conditions $r_0 = 2$ and $r_1 = 10$.

Find c and d .
